Mine your data: contrasting data mining approaches to numeric and textual data sources

IASSIST May 2006 conference

Ann Arbor, USA

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Data and text Mining

- Data mining is the exploration and analysis of large quantities of data in order to discover meaningful patterns and rules
- Typically used in domains with structured data, e.g. customer relationship management in banking and retail
- Text mining – extracting knowledge that is hidden in text to present distilled knowledge to users in a concise form
- Can collect, maintain, interpret, curate and discover knowledge

Data Mining

Data Mining originated in 90's as Knowledge Discovery or KDD

Knowledge Discovery in Databases

"world of networked knowledge"

- Directed data mining

  a variable (target) is explained through a model

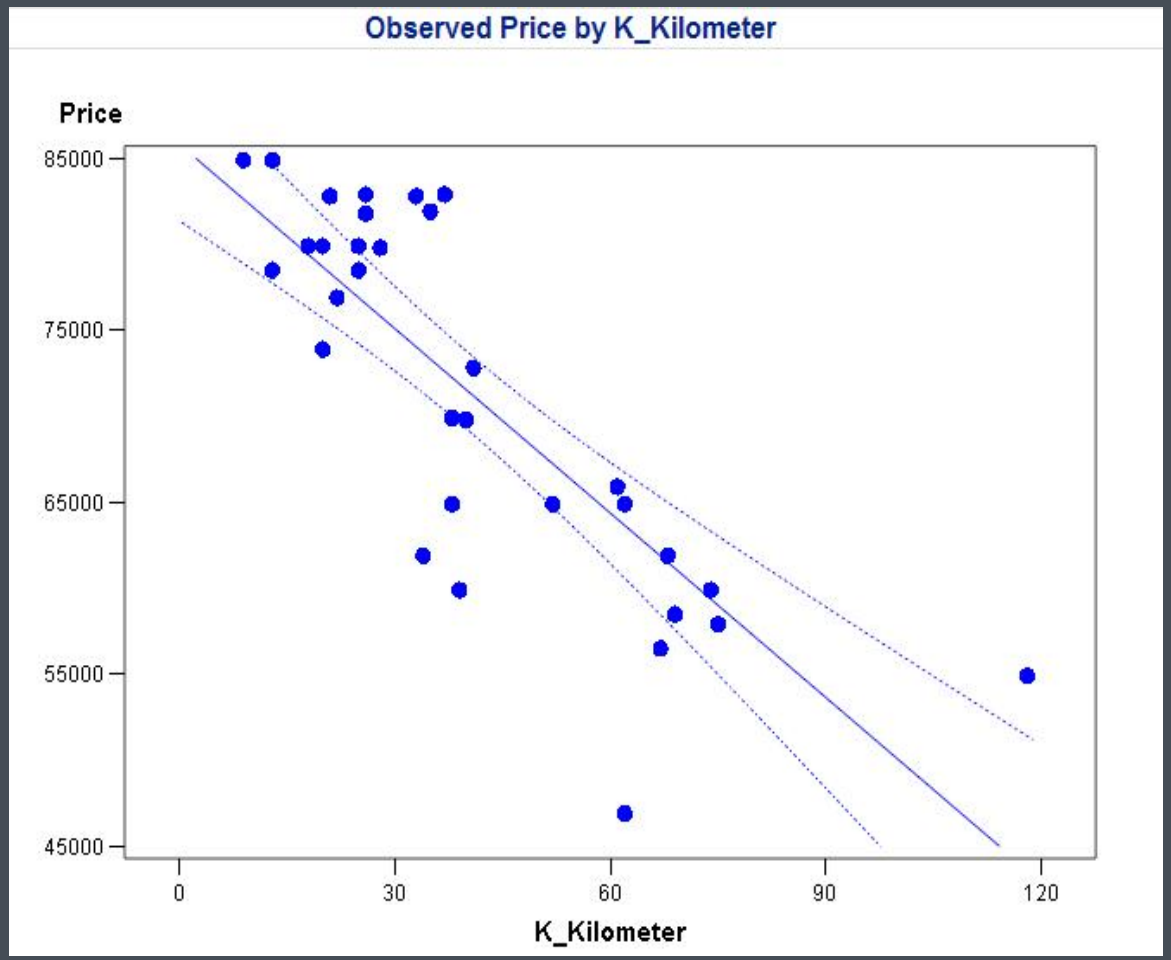
Model & Meaning

- "Meaning" may be regarded as an approximate synonym of pattern, redundancy, information, and "restraint"
- Knowing something
- "It is possible to make a better than random guess"

Bateson

Regression – visualization of the model

Used Nissan cars of same type: price, driven kilometers, year, color, paint, rust, bumps, non-smoking, leather, etc.



Regression - Model

- Linear

$Y = \alpha + \beta_1 X_1$

$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \ldots$ M[illegible]ependent variables

- Logistic

logit([illegible]g(P/(1-P)) = $\alpha + \beta_1 X_1$

[illegible]p($\alpha + \beta_1 X_1$) / (1 + ex[illegible] $\beta_1 X_1$))

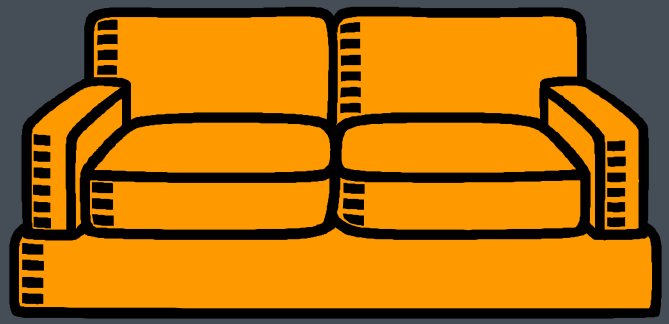
$P = \exp^{\alpha + \beta_1 X_1} / (1 + \exp^{\alpha +}$ [illegible]

- Quadratic .. etc.

The target & the problem

- Context: Selling via mail or e-mail or phone or.... directed towards a person

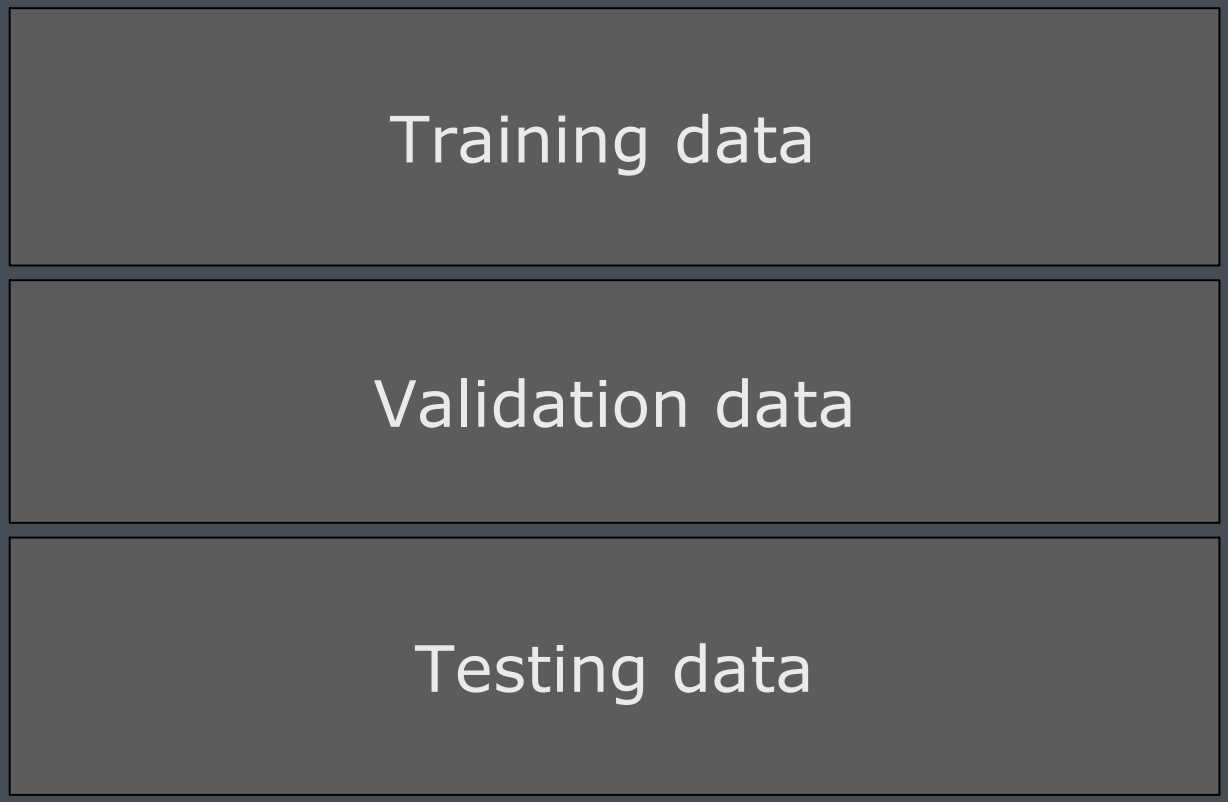
- We know the previous customers (potential customers) and which of these that bought our target



- Problem: we have 390 sofas to sell !

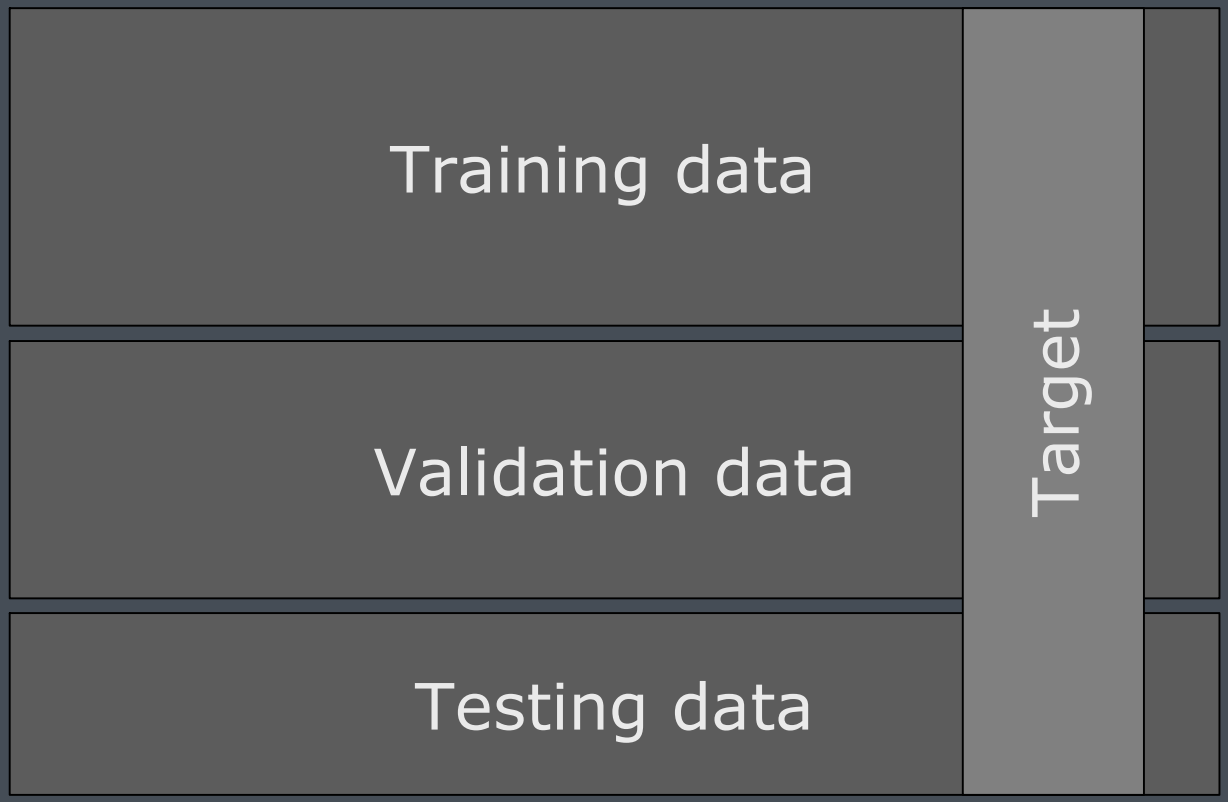
Lots of other models - and lots of data

- Split up the huge dataset



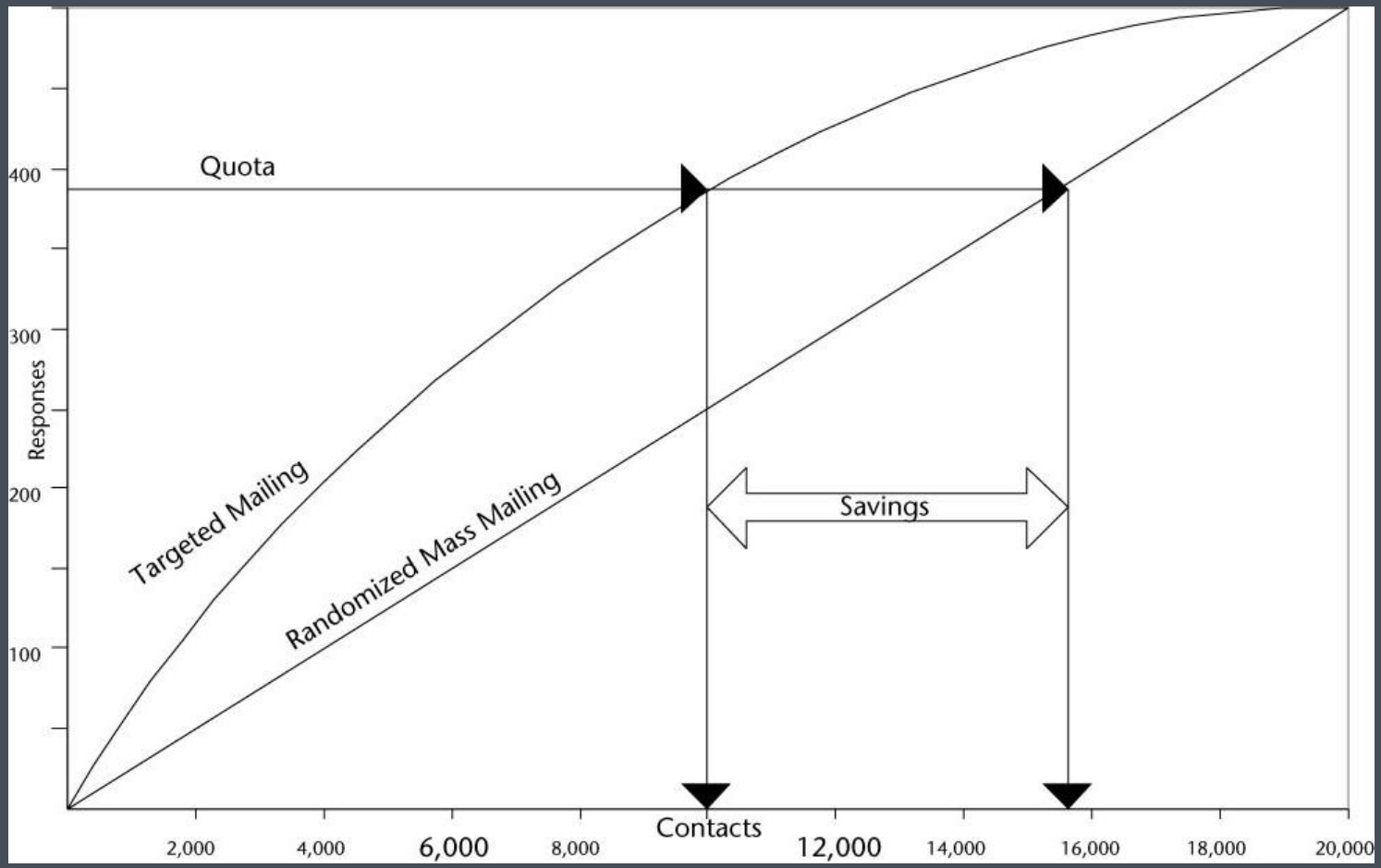
Lots of data

- Split up the huge dataset - random distributed



Ranking Prospects after the target

-



Confusion Matrix – we do make errors

But we use data with known outcome

	True Sale (positive)	True Non-Sale (negative)
Predicted Sale (predicted)	true positive	false positive
Predicted Non-Sale	false negative	true negative

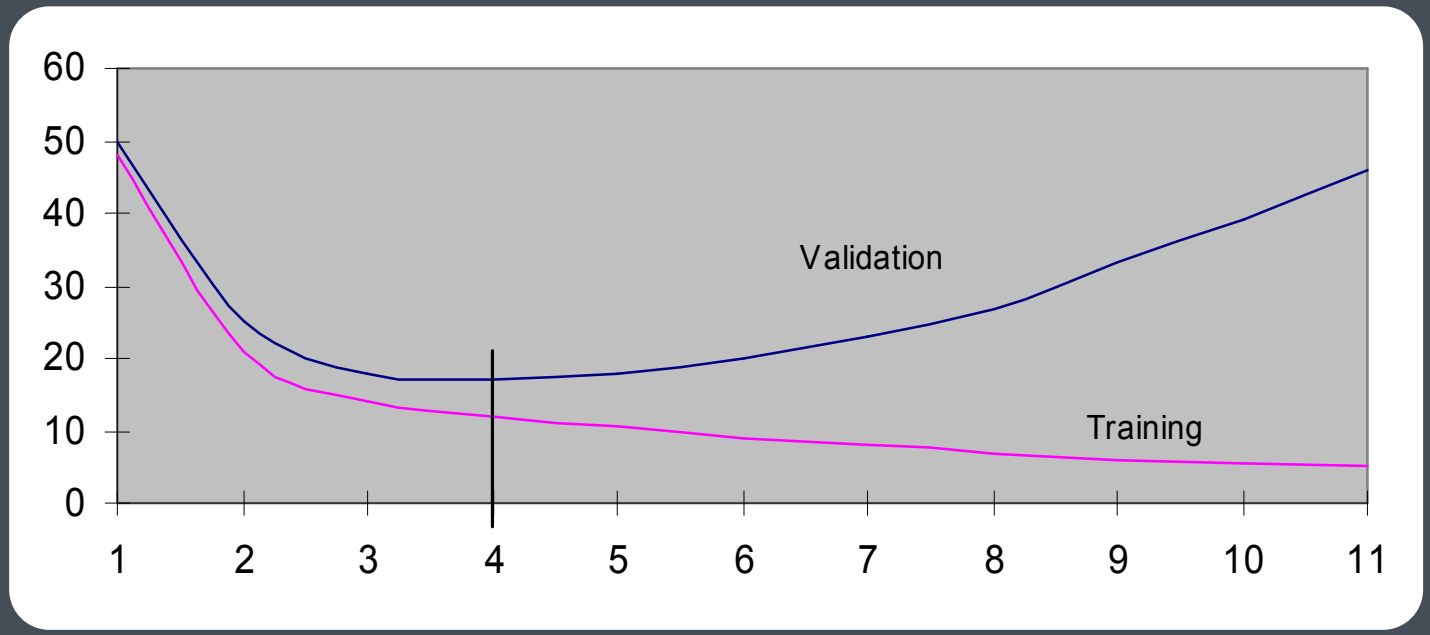
Error rate: rate of misclassification (false / all)

Sensitivity: prediction of true occurence (true positive / positive) (Recall)

Specificity: prediction of non-occurrence (true negative / negative)

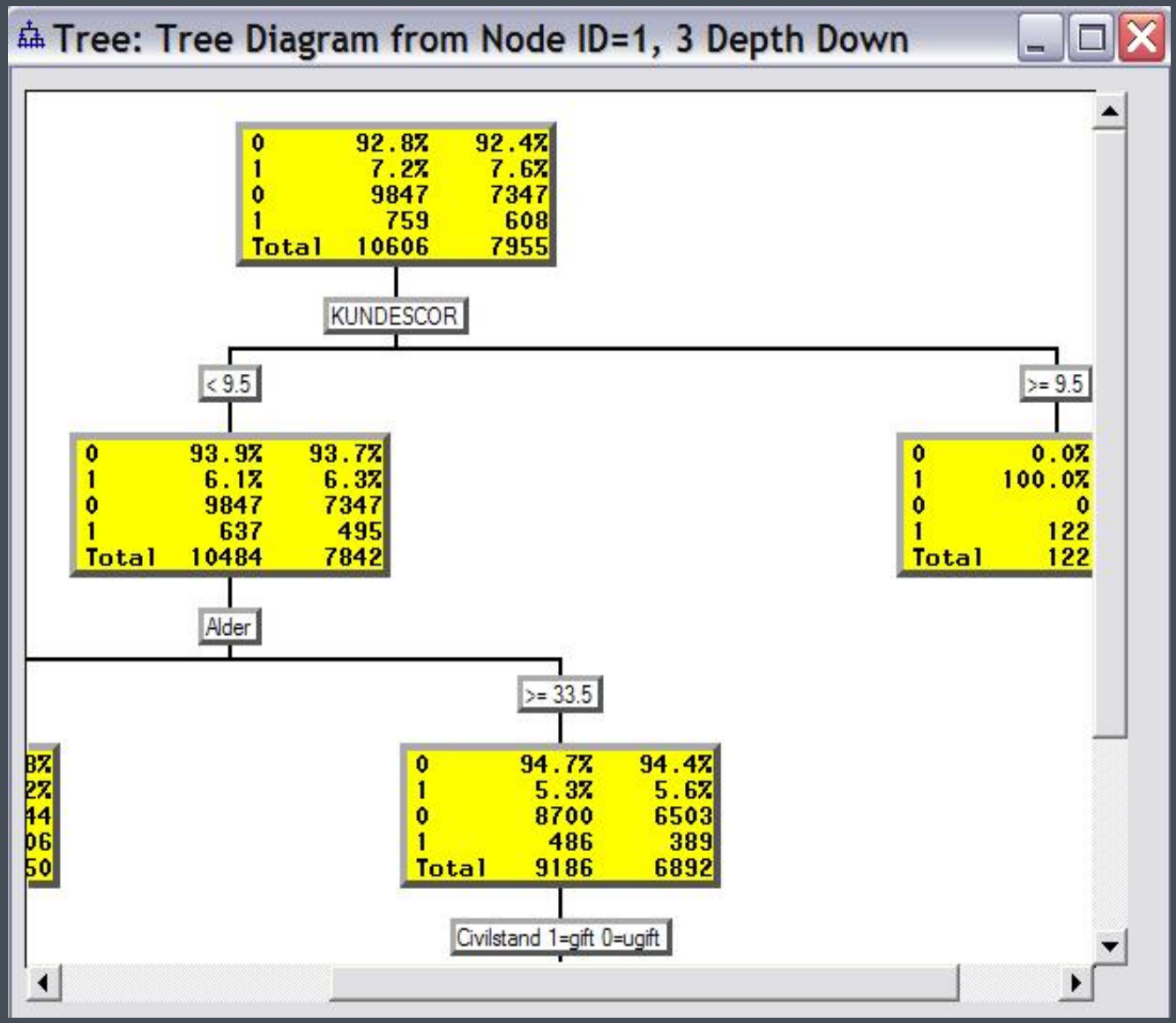
Precision: the truth in the prediction (true positive/predicted)

Overfitting

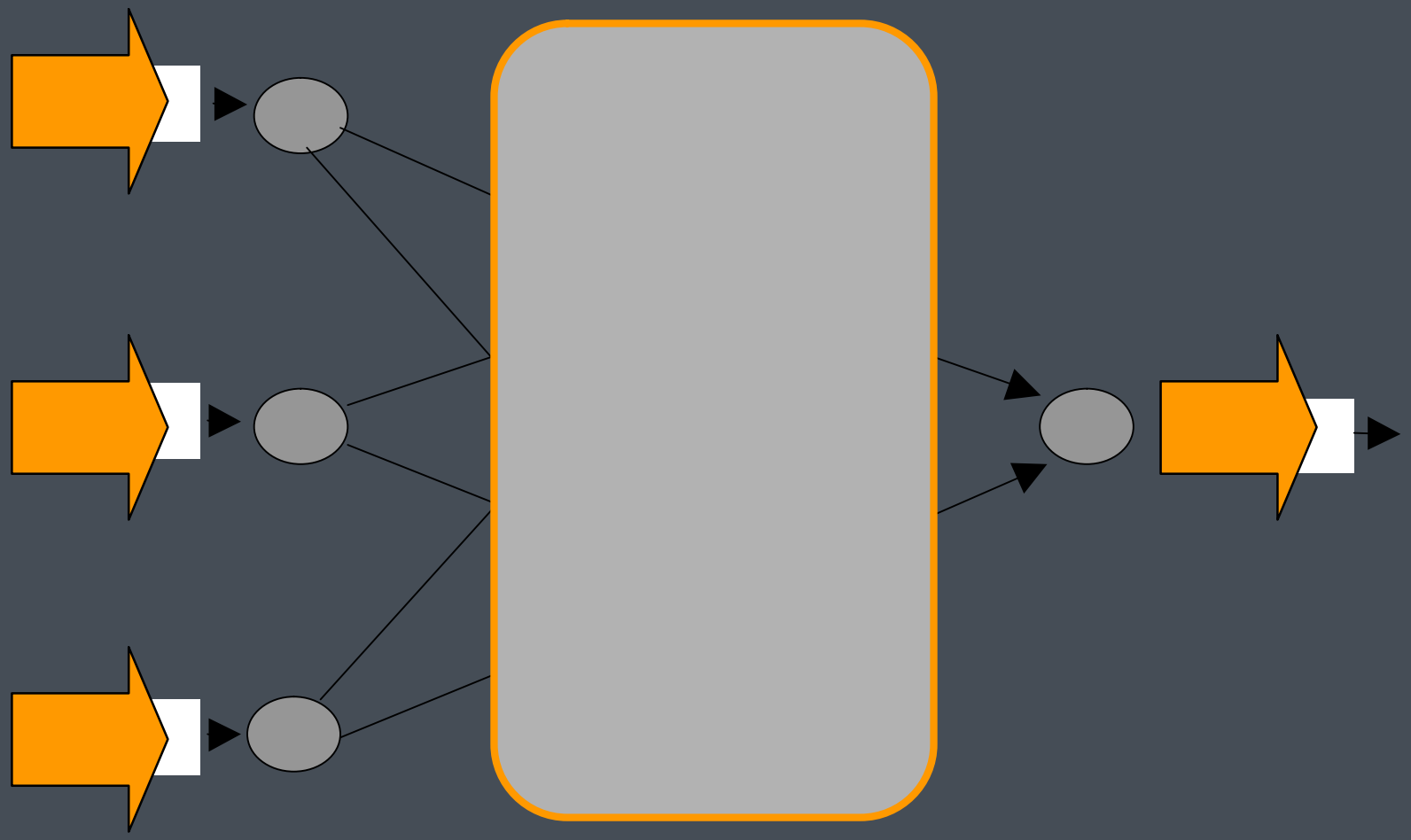


- Error rate after iterations

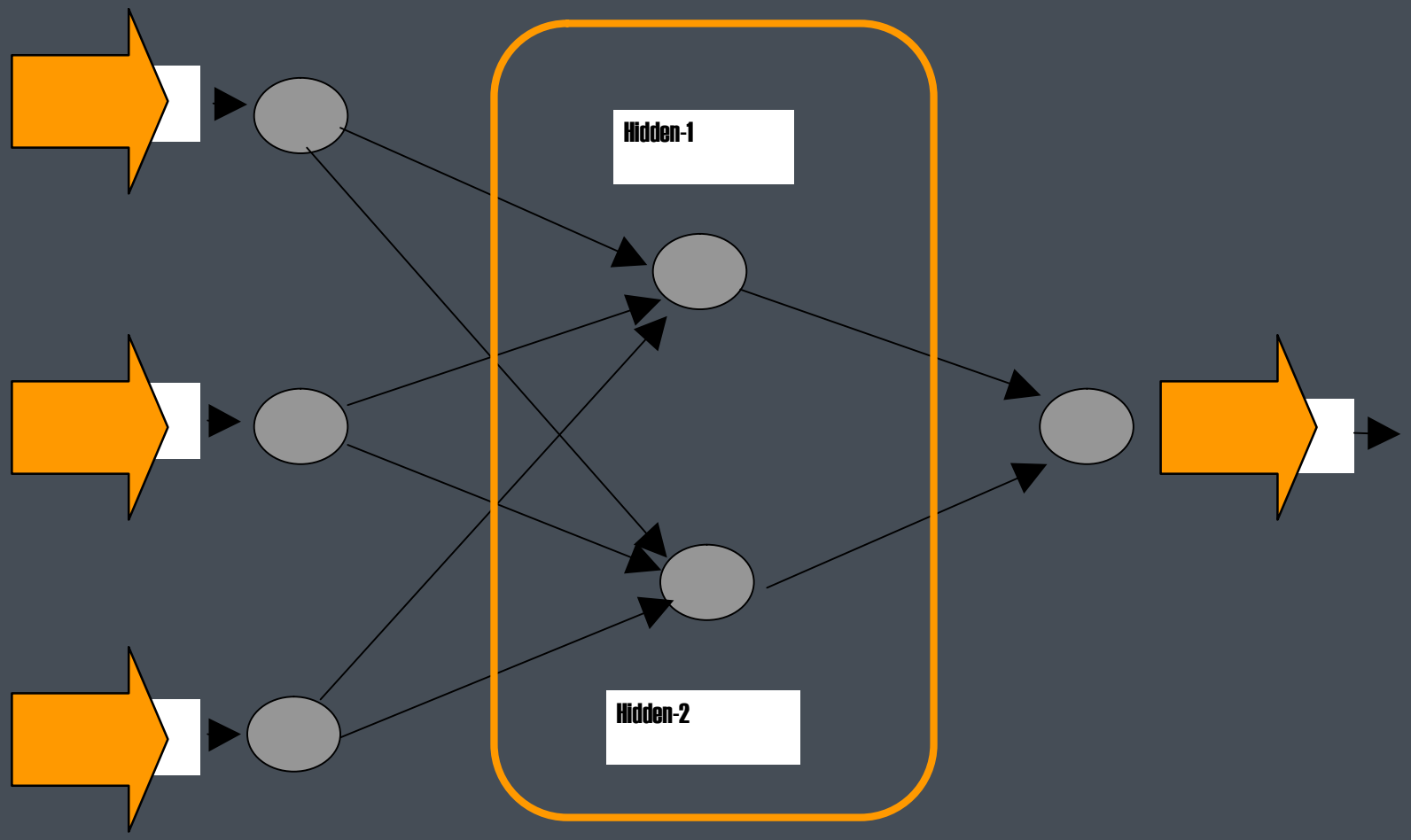
Another model – the Tree



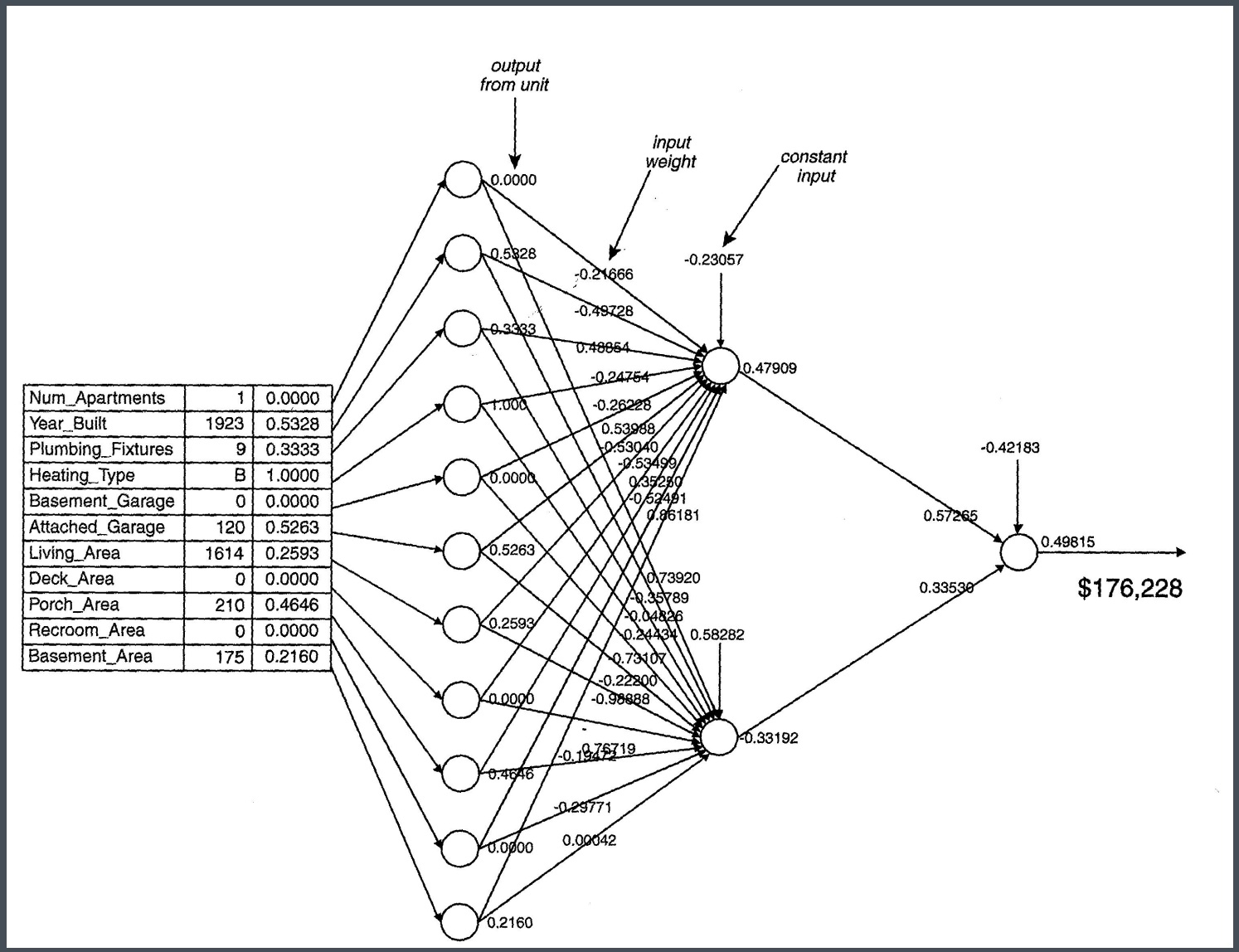
Neural network



Neural network – hidden layer



Weights in the neural network



Num_Apartments	1	0.0000
Year_Built	1923	0.5328
Plumbing_Fixtures	9	0.3333
Heating_Type	B	1.0000
Basement_Garage	0	0.0000
Attached_Garage	120	0.5263
Living_Area	1614	0.2593
Deck_Area	0	0.0000
Porch_Area	210	0.4646
Recroom_Area	0	0.0000
Basement_Area	175	0.2160

Comparing Models

Knowledge in a pragmatic way

- Using the model that works !
- Does not always know why it works !
- Nor for how long - forever is a long time
- And don't know what to look out for

- Good exploration leads to
  theory, hypothesis testing, etc.
- Demand for huge dataset in all dimensions

From analysis of well structured data

- We have experience and expertice!

	IP	Tidspunkt		F6	F7	F8	Element
101	200.37.136.13	03aug2003:05:11:25	+0	GET /latest/picture/?locationID=&partnerAc	200	2121	/latest/picture/?loca
102	200.37.136.13	03aug2003:05:11:25	+0	GET /latest/picture/?locationID=&partnerAc	200	2150	/latest/picture/?loca
103	200.37.136.13	03aug2003:05:11:26	+0	GET /latest/text/?locationID=&partnerAccou	200	199	/latest/text/?locatio
104	217.116.227.122	03aug2003:05:11:27	+0	GET / HTTP/1.0	302	203	/
105	200.37.136.13	03aug2003:05:11:27	+0	GET /latest/picture/?locationID=&partnerAc	200	2233	/latest/picture/?loca
106	200.37.136.13	03aug2003:05:11:27	+0	GET /latest/text/?locationID=&partnerAccou	200	197	/latest/text/?locatio
107	200.37.136.13	03aug2003:05:11:29	+0	GET /latest/picture/?locationID=&partnerAc	200	2086	/latest/picture/?loca
108	200.37.136.13	03aug2003:05:11:30	+0	GET /latest/picture/?locationID=&partnerAc	200	2492	/latest/picture/?loca
109	200.37.136.13	03aug2003:05:11:31	+0	GET /latest/text/?locationID=&partnerAccou	200	198	/latest/text/?locatio
110	200.37.136.13	03aug2003:05:11:32	+0	GET /latest/text/?locationID=&partnerAccou	200	202	/latest/text/?locatio
111	217.116.226.177	03aug2003:05:11:38	+0	GET /cron/parseMatrixes.php HTTP/1.1	200	5	/cron/parseMatrixe
112	200.37.136.13	03aug2003:05:12:36	+0	GET /browse/?id=0&SID=e30c9d51ac8572	200	7678	/browse/?id=0&SID
113	200.37.136.13	03aug2003:05:12:39	+0	GET /gfx/autogen/header.php?headerstring	302	5	/gfx/autogen/heade
114	200.37.136.13	03aug2003:05:12:39	+0	GET /inc/matrix.js HTTP/1.1	200	1021	/inc/matrix.js
115	200.37.136.13	03aug2003:05:12:39	+0	GET /misc/sessionStarter/index.php?backU	302	5	/misc/sessionStart
116	200.37.136.13	03aug2003:05:12:40	+0	GET /gfx/autogen/header.php?headerstring	200	1062	/gfx/autogen/heade
117	200.37.136.13	03aug2003:05:12:41	+0	GET /inc/matrix2.js?cardType='kioskCard'&l	200	6645	/inc/matrix2.js?card
118	200.37.136.13	03aug2003:05:12:44	+0	GET /latest/picture/?locationID=&partnerAc	200	1914	/latest/picture/?loca
119	200.37.136.13	03aug2003:05:12:44	+0	GET /latest/text/?locationID=&partnerAccou	200	201	/latest/text/?locatio
120	200.37.136.13	03aug2003:05:12:45	+0	GET /latest/picture/?locationID=&partnerAc	200	1914	/latest/picture/?loca
121	200.37.136.13	03aug2003:05:12:47	+0	GET /latest/text/?locationID=&partnerAccou	200	199	/latest/text/?locatio

To analysis of unstructured data

- Most information is semi-structured
- texts: e-mails, letters, documents, call-center, web-pages, web-blogs, ...

1,192 Terms

☐ Display dropped terms
☑ Display kept terms

Filter | Find Similar

Term	Freq	# Documents	Keep	Weight	Role	At
in	217	186	Y	0.252	Prep	Al
+ customer	148	140	Y	0.286	Noun	Al
with	78	75	Y	0.376	Prep	Al
+ company	70	68	Y	0.390	Noun	Al
on	70	65	Y	0.399	Prep	Al
+ bank	69	61	Y	0.410	Noun	Al
+ market	67	60	Y	0.412	Noun	Al
new	61	55	Y	0.424	Adj	Al
+ service	53	52	Y	0.428	Noun	Al
+ insurance	56	50	Y	0.438	Noun	Al
ref.	49	49	Y	0.436	Abbr	A
will	44	43	Y	0.456	Aux	Al
+ business	43	41	Y	0.464	Noun	Al
+ make	44	40	Y	0.469	Verb	Al
's	38	37	Y	0.478	Part	U
+ have	37	33	Y	0.500	Verb	Al
+ employee	33	32	Y	0.499	Noun	Al

Clusters

Filter | Find Similar

#	Descriptive Terms	Freq	Percentage	RMS Std.
1		343	35%	0
2	+ insurance, + market, + business, on, + area, + product	112	11%	0.0956969161
3	+ loan, + good, + create, + customer, interest, + offer	53	5%	0.0927655624
4	+ reduce, + cost, + director, new, + board, + reduction	38	4%	0.0861319145
5	secure, between, can, + future, ref., + product	83	8%	0.0971627534
6	+ service, + increase, + level, + improve, + organization, + provide	68	7%	0.0919957198
7	number, + employee, management, + branch, other, + expand	143	14%	0.0982256427
8	trade, + company, in, unibank, danish, + investment	146	15%	0.0962443483

Structure in text

IASSIST 2006 - Mozilla Firefox

File Edit View Go Bookmarks Tools Help

http://www.icpsr.umich.edu/iassist/

Getting Started Latest Headlines

IASSIST 2006

Data in a World of Networked Knowledge

- Front Page
- Contact Us

Logistics

- Registration
- Arrival instructions
- Accommodations
- Visitor information
- International Delegates
- Maps

Meeting Events

- Call for papers
- Meeting program
- Workshops
- Conference poster (PDF 17MB)
- Poster session
- Tuesday reception
- Thursday banquet
- Michigan excursions
- Sponsors

May 23-26, 2006 - Ann Arbor, Michigan

Welcome to the conference Web site for IASSIST (International Association for Social Science Information Services & Technology) 2006. The 2006 meeting will be jointly hosted at the University of Michigan in Ann Arbor by the Inter-university Consortium for Political and Social Research (ICPSR), the University of Michigan Libraries, and the School of Information. Note that the IASSIST conference venue usually follows a four-year cycle of USA, Canada, USA, and Europe.

The main conference begins Wednesday, May 24, and ends Friday, May 26. It will be preceded by a meeting of the IASSIST Administrative Committee on Monday and workshops on Tuesday, and then followed by a meeting of the DDI Alliance Committee on Saturday. Tours will also be offered on Saturday for those interested in seeing more of the Ann Arbor area and its attractions.

The workshops will take place concurrently at ICPSR, U-M's Science Library, and the School of Information. Meeting sessions will take place on the U-M campus at the Michigan Union.

Online registration is now available!

IASSIST 2006 Sponsors

IASSIST would like to thank the sponsors of the 2006 conference! Please visit the sponsor page for a list of 2006 sponsors and to learn more about the products and services they provide.

Done

Text mining

Extracting precise facts from a retrieved document set or finding associations among disparate facts, leading to the discovery of unexpected or new knowledge

Activities

- Terminology management
- Information extraction
- Information retrieval
- Data mining phase –find associations among pieces of information of extracted information

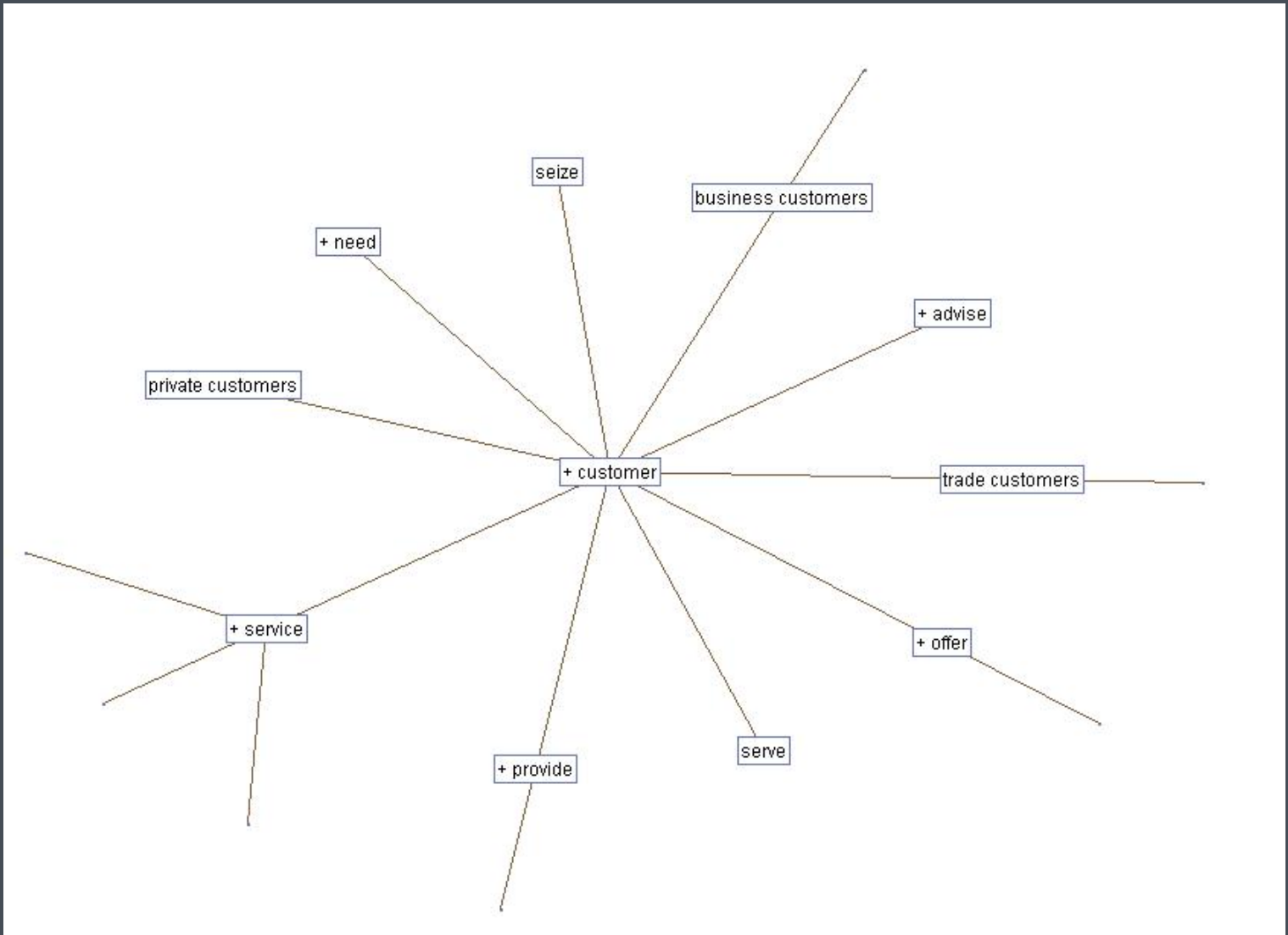
How can text mining help?

- Distill information
- Extract ‘facts’
- Discover implicit links
- Generate hypotheses

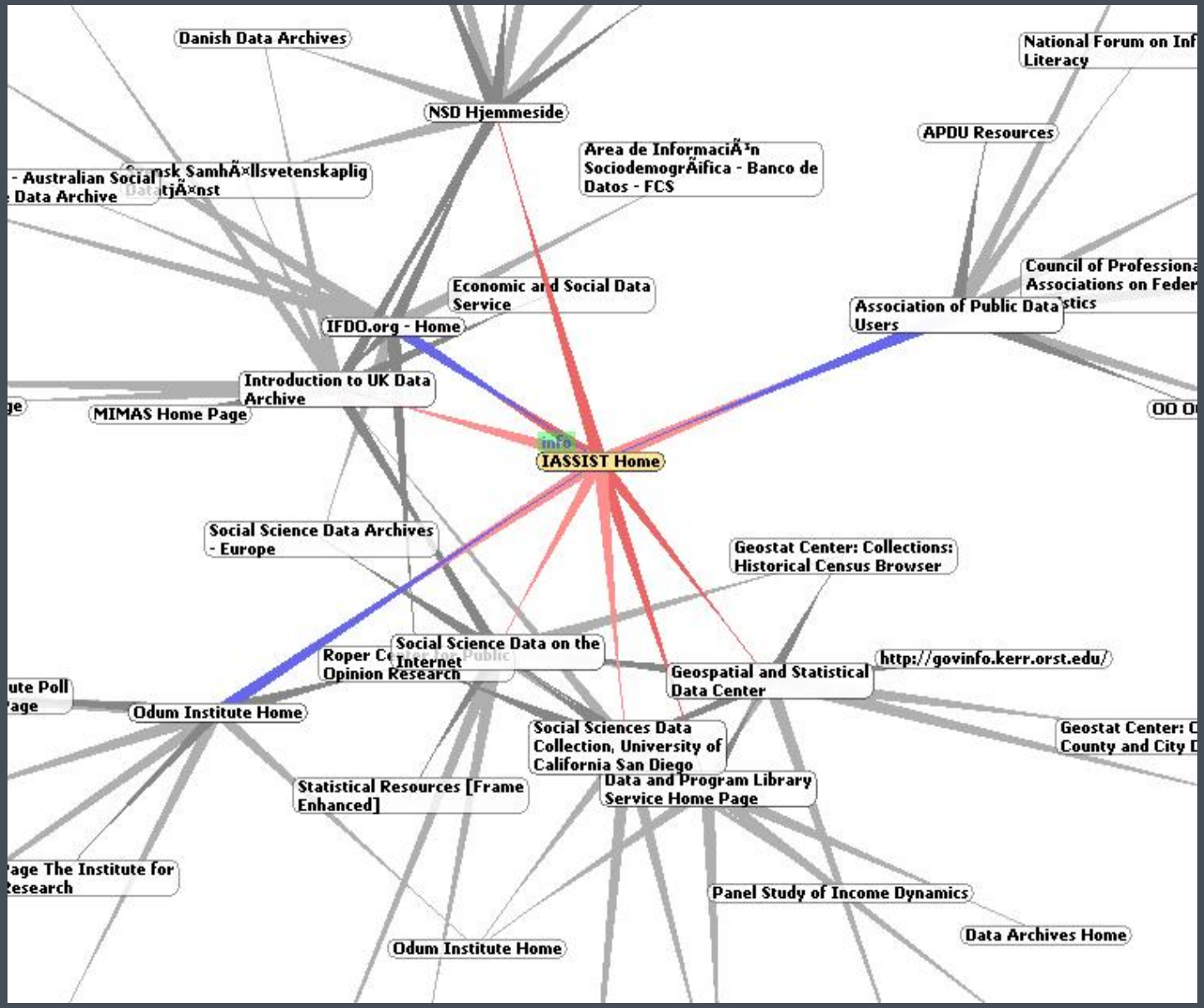
Entities and concepts

- Extraction of named entities
  - People, places, organisations, technical terms
  - Discovery of concepts allows semantic annotation of documents
  - Improves information by moving beyond index terms,
  - Enabling semantic querying
- Can build concept networks from text
  - Clustering and classification of documents
  - Visualisation of knowledge maps

Knowledge map



Visualizing links



Popular fields for text mining

Applicable to science, arts, humanities but most activity in:

- biomedical field
  - identify protein genes e.g. search whole of Medline for FP3 protein activates/induces enzyme
- government and national security – detection of terrorist activities
- financial – sentiment analysis
- business – analysis of customer queries/satisfaction etc

Text mining tasks and resources

- Documents to mine
  - texts, web pages, emails
- Tools
  - parsers, chunkers, tokenisers, taggers, segmentors, entity classifiers, zoners, annotators, semantic analysers
- Resources
  - annotated corpora, lexicons, ontologies, terminologies, grammars, declarative rule-sets

Example: speech tagging

- input document with word mark-up

- apply tagging tool

- output additional mark-up of part of speech

Example: named entity tagging

Mozilla Firefox

File Edit View Go Bookmarks Tools Help

U id='1' who='interviewer' Right, it starts with your grandparents. So give me the names and dates of birth of both. Do you remember those sets of grandparents?

U id='2' who='subject' Yes.

U id='3' who='interviewer' Well, we'll start with your mum's parents? Where did they live?

U id='4' who='subject' They lived in Widness, Lancashire.

U id='5' who='interviewer' How do you remember them?

U id='6' who='subject' When we Mum used to take me to see them and me Grandma came to live with us in the end, didn't she?

U id='7' who='Welham' Welham: Yes, when Granddad died - '48.

U id='8' who='interviewer' So he died when he was 48?

U id='9' who='Welham' Welham: No, he was 52. He died in 1948.

U id='10' who='interviewer' But I remember it. How old would I be then?

U id='11' who='Welham' Welham: Oh, you would have been little then.

U id='12' who='subject' I remember him, he used to have whiskers. He used to put me on his knee and give me a kiss.

U id='13' who='Welham' Welham: Well, he'd have whiskers where he probably had not had a shave and he used to rub her cheek - literally.

U id='14' who='interviewer' What was his occupation?

U id='15' who='Welham' Welham: He worked at ICI, Widness.

U id='16' who='interviewer' Doing what?

U id='17' who='Welham' Welham: What did he do at ICI? Worked at ICI.

U id='18' who='interviewer' What was Grandmother's occupation?

U id='19' who='Welham' Welham: She had nought, because she had arthritis. She couldn't walk. That's why I used to go home every weekend to look after her. We had a Home Help until Friday and I used to go Saturday and stop until Monday when the Home Help came again.

U id='20' who='interviewer' Really? This was when you were married?

U id='21' who='Welham' Welham: Yes. I took me children with me. Me husband, give him his due, he let me go every Saturday, tea-time, until Monday and I used to take the children with me.

U id='22' who='interviewer' And then she came to live with you eventually?

U id='23' who='Welham' Welham: Well, me Dad died. Me sister was younger than me and she got TB and she went in Rufford Sanatorium but I still come every week and looked after me Mum on a Saturday night. Got up, got the dinner ready and got the bus from Widness to Rufford every Sunday to see me sister. Well, she was in for two years in Rufford and then they said they couldn't do nothing, so I brought her home and I nursed her for six weeks and she died and that's when we Mum came to me and we give the house up. There was nothing left. I got three brothers, but wives are not the same as their own daughter. So we all fell out over who was going to have her. So I said well, nobody's having her only me. I brought her to Altrincham and she lived two years. I made her comfortable in the kitchen with a single bed and looked after her. Gave me little job up and I looked after her and she just lived two years.

Key: organisation person location date time money percent geo-political vehicle facility weapon

Done | Radar: | Now: Partly Cloudy, 8° C | Fri: 3° C | Sat: 7° C

Document clustering

- information retrieval systems based on a user-specified keyword can produce overwhelming number of results
- want fast and efficient document clustering – browse and organise
- unsupervised procedure of organising documents into clusters
  - hierarchical approaches (partitional)
  - K-mean variants
- terminological analysis based on extracted documents to identify named entities, recognise term variations
- perform query expansion to improve the recall and precision of the documents retrieved

Processing steps

- submit abstracts
- filter by
  an ontology
  applying criteria - date, language, author, no data reported
- include or exclude documents
- cluster by ranking
- auto summarise using ‘viewpoints’
  Use full parsing and machine learning techniques
- apply to test annotated corpus
- output relevant extracted sentences

Automatic document summarisation

- Document Understanding Conferences (DUC)
- Message Understanding Conferences (MUC)
- Text Summarisation Challenge (TSC)

Groups undertake specified concrete tasks to generate summaries based on set queries

1. Input our extracted sentences
2. Summarise into subsections by topic
3. Extract salient information
4. Exclude redundant information
5. Maintain links from summaries to the source documents

Social science and text mining

- in UK text mining not been applied to social science data - to published reports nor raw data
- two realistic social science applications:
  - ✓ helping with new field of 'systematic review' of social science research from published abstracts
  - ✓ helping 'process' (enrich) shared qualitative data sources for web publishing and sharing
- both relatively new fields – last 10 years
- UKDA and Edinburgh/Manchester/Essex NLP and text mining connections are a first in UK/Europe

Limitations of basic NLP tools

- plethora of tools across institutes
- many tools are individually honed for specific purposes e.g. biomedical applications
- often tools and output from tools are non-interoperable - hard to bolt components together
- NLP tools are ugly – unix/linux command-line programs communicate via pipes
- often useful to draw on range of existing tools for different processing purposes

Text mining services

- Centre for Text Mining in the UK
  - develop tools - demonstrators
  - processing service with packaging of results
  - best practice, user support and training
  - access to ontology libraries
  - access to lexical resources – dictionaries, glossaries and taxonomies
  - data access, including annotated corpora
  - grid based flexible composition of tools, resources and data ..portal and workflows

The power of the GRID

- at present, social science problems have typically not required huge computational power
- computational power is needed for undertaking large-scale data and text mining
- searching for a conditional string across millions of records can take hours
- data grid useful for exposing multiple data sources in a systematic way using single sign on procedures

Mining and the GRID

- parallel power
  - distribute processes over lots of machines
  - use parallel algorithms to speed up processing tasks
- access to distributed data and models
  - multiple pre-processed textual data
  - distributed annotation of text
  - models with provenance metadata
- processing pipeline distributed
  - tools/components are hosted at different sites
- but what about curation, exposure and systematic description of data sources?

Challenges for mining

- maximise the interoperability of processing resources
- maximise shared data and metadata resources in a distributed fashion
- enable simplified yet safe sharing and respect for ownership
- innovative methods of visualisation
- hide any nasty behind the scenes business from the ‘average user’ (processing programs, authentication middleware etc)
- New Web Services, registries, resource brokers, and protocols
- juggling data dimensions from atomic data to aggreggations

- Thanks

- Louise Corti & Karsten Boye Rasmussen